CogniBase

BU-Aligned Document Library

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Contents

CogniBase — Executive Brief & BU Vision Alignment

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1. In one paragraph

Boston University has made a deliberate architectural bet: AI and institutional data are two halves of one problem, and the answer is an owned, open, semantically-governed platform — Atlas — in which every agent reasons through an **ontology**, a **semantic layer**, and a **curated knowledge base** before it ever touches a raw system. **CogniBase brings Hyland OnBase into that discipline.** It is the OnBase-native member of the intelligence stack: it maps the schema, governs the corpus, resolves entities on sanctioned relationships rather than coincidence, and answers natural-language questions across OnBase and adjacent institutional data — with every cross-source claim auditable back to the rule that authorized it. It speaks BU's own fabric (MCP tools, OPA policy, Entra identity, an L1–L4 audit trail), so it is not a bolted-on product but a governed node BU controls.

2. BU's direction, stated plainly

BU's published posture is a **"critical embrace" of AI**, operationalized as three commitments and a target architecture:

BU commitment	What it means	How CogniBase honors it
Architecture first	Build the foundation (ontology, semantic layer, owned knowledge) before scaling apps	CogniBase is a foundation layer for OnBase, not a chat skin over it
Curated heterogeneity	Many models/platforms, no single bet; each earns its place on fit, cost, capability, *freedom to leave*	Vendor-pluggable LLMs (Claude/GPT/Gemini/local); plain-file artefacts; open exit
AI-native, not AI-assisted	Rebuild workflows so AI is the core interface	OnBase content becomes *queryable as knowledge*, not navigated as documents

BU's **target intelligence stack** — the spine of the whole program — is four layers every agent must reason through, *never reaching raw systems directly*:

Agent → Knowledge base → Semantic layer → Ontology

This brief's thesis: **OnBase is the largest institutional content store not yet represented in that stack — and CogniBase is how it gets there, correctly.**

3. The missing node

BU's current grounding sources — Person, Course, Student Hub, WordPress, ServiceNow — are real but partial. The university's system of record for contracts, invoices, transcripts, HR files, and decades of imaged documents is **OnBase**, and OnBase is uniquely hard to bring into an AI platform:

- **2,300+ tables, cryptic names, and almost no enforced foreign keys** — relationships live in application logic and configuration packages, not in the database.
- **Regulated content** (FERPA, GLBA) that cannot be casually shipped to a cloud model.

- **A configuration domain** (DocTypes, KeywordTypes, Lifecycles, Custom Queries) whose *meaning* is invisible to a naïve "chat with your database" tool.

CogniBase was built specifically for this. It does not displace anything on BU's roadmap — it fills the empty OnBase slot with a component designed to BU's own rules.

4. The trust thesis (why this brief leads with ontology)

The defining risk of putting an AI over a data lake is not that it fails to find correlations — it is that **it finds too many, convincingly, that are not real.** A model with access to data but not to the *business-process model* behind it will assert that an OnBase `AP_Name` "J. Smith," an SIS student "John Smith," and an HR record are the same entity — a fluent, fully-cited analysis that an audit later proves false. A neophyte believes it; the institution acts on a fiction.

CogniBase's central design commitment is the opposite stance: **a correlation is trustworthy only if a governed ontology sanctions it.** The system traverses only declared relationships, resolves entities on vetted keys (never name or embedding similarity alone), and labels anything unsanctioned as *inferred and unverified* — excluded from authoritative claims. This is the subject of Documents 2 (Semantic & Data Ontology) and 3 (Challenges & Mitigations), and it is the single most important reason CogniBase belongs in an *institutional* — not merely a clever — AI program.

5. How CogniBase maps onto Atlas

CogniBase is designed to be an Atlas-native citizen, not a foreign service:

Atlas component	CogniBase relationship
Nexus (AI platform, MCP spine)	CogniBase registers in the `mcp-gateway-registry`; Nexus/Regent agents call its tools (search, ask, schema-explain, federated-query)
Cortex (Iceberg/S3 lake, Trino)	Curated OnBase entities can be emitted to the lake — Trino-queryable, lineage-tracked
Lattice (APISIX, FastAPI)	CogniBase is FastAPI; deployable behind APISIX as a governed "OnBase API"
Lexicon (semantic layer, knowledge graph)	CogniBase's Normalizers/ontology feed BU's planned enterprise knowledge graph
Forge (EKS, OPA, Entra, audit)	Gates → OPA policy; identity → Entra ID; L1–L4 audit → BU's audit sinks

The integration surface is **MCP first** (lightest, native), with a data-plane API path as the durable Phase-2 home. Detail in Documents 4, 11, and 14.

6. What a pilot proves

A scoped pilot against a non-production OnBase instance demonstrates, in BU's own terms:

1. **Governed correlation** — a real cross-DocType / OnBase-↔-SIS question answered through *sanctioned* relationships, with the inferred-vs-verified distinction visible to the user.

2. **Auditability** — every number traceable to its source and the rule that joined it (FERPA-grade).

3. **Atlas-native integration** — CogniBase reachable as an MCP tool from a Nexus agent.

4. **No lock-in** — artefacts as plain files; the whole profile portable; open exit demonstrated, not promised.

7. What we ask of BU

Consistent with BU's "Working Together" model: a **sponsoring team**, a **named technical contact**, and **one IT credential request** for a shared non-production OnBase instance usable by both CogniBase and BU's REST POC team. Coordination, scope, and roadmap are detailed in Document 16.

8. Reading guide to this library

- **Trust layer (read first):** Doc 2 *Semantic & Data Ontology*, Doc 3 *Challenges & Mitigations*.
- **Architecture & knowledge:** Docs 4–10 (architecture, components, corpus, access, federation, hygiene, provenance).
- **Platform, model & delivery:** Docs 11–14 (MCP/Nexus, model strategy, deployment/licensing, API & tool contract).
- **Adaptive trust & learning:** Document 17 — *Confidence, Human-in-the-Loop Review & Continuous Learning*.
- **Systems landscape:** Document 18 — *BU Systems of Record: Affinity & Extension Map*.
- **Appendices:** Appendix A — *MapSnap × CogniBase Tandem & Pre-Ingestion Augmentation*; Appendix B — *Alignment to BU's 2026 Roadmap (Four Lanes)*.
- **Strategy & engagement:** Doc 15 *Beyond OnBase*, Doc 16 *Coordination & Pilot Plan*.

> *CogniBase's promise to BU is not "an AI that talks to OnBase." It is **OnBase made institution-grade knowledge** — governed, semantic, auditable, and owned — on the architecture BU has already chosen.*

CogniBase — Semantic & Data Ontology

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The trust layer: how cross-source correlation is made governed, not inferred.

1. The problem, stated precisely

Put a capable model over a data lake and the danger is **not** that it fails to find relationships — it is that it finds **too many, too convincingly, that are not real.**

Three things get silently conflated:

- **Co-location** — two datasets sit in the same lake.
- **Correlation** — two fields look statistically or semantically similar.
- **Real relationship** — two records describe the same business entity, joined on a key the business actually uses.

An AI that has the **data** but not the **business-process model** behind it treats co-location as correlation, and correlation as real relationship. The result is a **fluent, fully-cited, and wrong** analysis. It reads as authoritative because the language model is good at language — not because the join is valid.

> ***Why this is dangerous, not merely imperfect.*** *A specialist sees the error in seconds ("those aren't the same Smith"). A neophyte — or an executive reading a dashboard — cannot. They act on a fiction. Months later an audit reconstructs the join, finds no real connection, and the institution's confidence in every AI output collapses. One convincing-but-false correlation costs more trust than ten honest "I don't knows."*

1a. What this failure mode is called

This is not a novel risk — it is a **recognized failure mode** with established names across disciplines. Naming it precisely matters for a technical audience:

- **Spurious correlation** — an association that looks real but reflects no genuine relationship *(statistics)*.
- **Conflation** (entity conflation / over-merging) — treating two distinct real-world entities as one *(identity & records management)*; the "two Smiths" merge.
- **Fan trap / chasm trap** — a join path that returns plausible but **wrong** results because the data model does not reflect the real relationship *(data modeling)*.
- **Conflating correlation with causation** — mistaking association for a real or causal link *(causal inference)*.
- **Ungrounded inference / the symbol-grounding problem** — manipulating symbols without connecting them to real-world referents *(AI / cognitive science)* — the root cause.
- **Confabulation** — fluent, confident, unfounded output *(as applied to LLMs)*.

CogniBase's working term **fabricated correlation** is the umbrella for these. The deeper framing: **semantics and ontology supply the *"what"* and *"what it is called"*; what is missing is *pragmatics* — meaning in real-world context — and the *business-process / causal model* that licenses a relationship.** An ontology says a Vendor and an Employee *can* both be a Person; only the process and grounding establish that *this* vendor **is** — or, crucially, **is not** — *this* employee. As Korzybski put it, *the map is not the territory*: the model must never be mistaken for the institution it represents.

2. Root cause

The model is missing the layer that gives data meaning: **the ontology and the business process that produced the data.** A lake stores *what is*; it does not store *what relates to what, on which key, within which boundary, and why.* OnBase makes this especially acute — it has **almost no enforced foreign keys**, so even the database itself does not assert how its data connects. The relationships live in configuration, conventions, and human knowledge that the AI never sees.

So the AI improvises the missing layer — and improvisation, dressed in confident prose, is indistinguishable from knowledge to a non-expert.

3. The governing principle

> ***A correlation is trustworthy only if a governed ontology sanctions it.***

Everything below follows from this one stance. The system may **traverse only declared relationships**, must **resolve entities on vetted keys**, and must **label anything else as inferred and unverified** — visibly, and excluded from authoritative claims. Joins are **deny-by-default**: silence in the ontology means "not connected," not "connect them anyway."

4. The discipline — eight best practices (and how CogniBase implements each)

#	Best practice (industry)	What it prevents	CogniBase mechanism
1	**Ontology as a contract** — typed entities + explicitly sanctioned relationships (edges), each with keys, cardinality, and boundary	Free-form "everything can join everything"	The relationship layer + CrossLinks (admin-vetted joins)
2	**Entity resolution on vetted keys with recorded confidence** — never string/embedding similarity alone	"J. Smith" = "John Smith" = employee	Normalizers (declared semantic equivalence, e.g. `AP_Name ≡ HR_Name → canonical:PersonName`) + match-confidence
3	**Semantic layer** — one documented definition per business term	"Which 'active enrollment' did you mean?"	Shared definitions surfaced to the agent; aligns to BU's Lexicon/Cube
4	**Business-process binding** — encode the workflow that gives data meaning	Treating a document's existence as its significance	Lifecycle/Custom-Query awareness from OnBase `.expk` configuration
5	**Provenance on every cross-source claim** — cite the sanctioned rule that permits the join	Unaccountable, unreproducible answers	L1–L4 audit; every correlation cites its CrossLink/Normalizer
6	**Confidence ≠ correctness guardrail** — surface	The neophyte over-trusting fluent output	Inferred results visibly labeled, withheld from

	evidence and boundaries; mark sanctioned vs inferred		numeric/authoritative claims
7	**Human-in-the-loop stewardship** — a domain owner ratifies relationships	Auto-discovered joins becoming silent fact	Auto-discover → **admin-confirm** workflow; steward sign-off
8	**Data contracts** — each source declares schema, semantics, keys; changes are versioned	Silent drift breaking joins	Per-source contracts + multi-version anchors in the hygiene pipeline

The throughline: **the ontology is the allow-list.** The AI's creativity is welcome in *language* and *hypothesis generation* — never in *asserting that two records are the same thing.*

5. Pragmatic BU examples (OnBase + the institutional sources)

Each example shows the **seductive-but-wrong** path, the **governed** path, and **what the audit sees.**

Example 1 — "Which vendors are also employees?" (conflict-of-interest)

Sources: OnBase AP DocTypes ⇄ HR/Finance (SIS/SAP).

- **Wrong:** match `AP_VendorName` to `HR_EmployeeName` on string/fuzzy similarity → "47 vendors are employees." Several are coincidental name matches; at least one merges two different people. Convincing, cited, **false.**
- **Governed:** the only sanctioned key is a **validated person identifier** (tax ID / BU person ID via the Person CDM). A Normalizer declares `AP_TaxID ≡ HR_TaxID → canonical:PersonTaxID`; the CrossLink authorizes the AP⇄HR join *only on that key*. Name is used to *display*, never to *join*. If a record lacks the key, it is returned as **inferred — unverified**, explicitly flagged, and excluded from the count.
- **Audit sees:** N matches on PersonTaxID (CrossLink #12), K inferred-only candidates listed separately with the reason "no shared validated key."

Example 2 — "Students with financial-aid documents stuck > 7 days, and their enrollment status"

Sources: OnBase FA DocType + Lifecycle ⇄ SIS.

- **Wrong:** join OnBase docs to SIS students on name or on an OnBase keyword that *looks* like a student ID but is a legacy applicant number → wrong students, wrong status.
- **Governed:** sanctioned key = **BU student ID** (Normalizer maps the OnBase keyword to `canonical:StudentID`); **boundary** = current term; **business-process binding** = the FA Lifecycle state ("In Review") defines "stuck," not the mere existence of a document. The Query Federator runs one sub-query per source and joins on StudentID only.
- **Audit sees:** the lifecycle state used, the term boundary, and the StudentID join — reproducible.

Example 3 — "Donors who are alumni and have transcripts on file" (multi-hop, FERPA)

Sources: Advancement ⇄ SIS ⇄ OnBase transcript DocType.

- **Wrong:** a 3-way fuzzy join across donor name, alumni name, and transcript keyword → a privacy incident waiting to happen and likely mis-linked people.
- **Governed:** each **edge** must be independently sanctioned — Advancement⇄SIS on `canonical:PersonID`, SIS⇄OnBase on `canonical:StudentID` — and a **Gate** enforces that **transcript *content** *never enters the answer (FERPA): the system may confirm* existence *and* count*, not disclose contents, unless policy is explicitly raised with a recorded agreement.
- **Audit sees:** two sanctioned edges, a privacy Gate decision, and a content-suppression record.

Example 4 — "HVAC service tickets vs. open facilities work orders for a building"
Sources: ServiceNow ⇄ OnBase Facilities DocType.

- **Wrong:** join on the free-text **building name** ("Photonics" vs "Photonics Center" vs "8 St Mary's") → split or merged buildings.
- **Governed:** the entity is the **asset/building**, resolved on a **facilities asset ID**; a Normalizer maps both sources' identifiers to `canonical:AssetID`. Building name is display-only.
- **Audit sees:** the AssetID join and the assets that had no sanctioned ID (reported, not guessed).

6. A pragmatic ontology approach for BU (don't boil the ocean)

BU's roadmap already lists an **"Ontology Investigation"** and a 12-month goal of an **enterprise knowledge graph**. CogniBase makes that incremental and evidence-driven for OnBase:

1. **Seed entities from what already encodes meaning** — the OnBase **`.expk` configuration** (DocTypes, KeywordTypes, Lifecycles) and the **Person CDM** (400+ attributes). These are pre-vetted semantic anchors, not guesses.

2. **Auto-discover candidate Normalizers** — propose `KeywordType ≡ KeywordType → canonical:X` equivalences; present them to a **steward to confirm or reject.** Nothing becomes fact without sign-off.

3. **Build the graph edge by edge** — each CrossLink is a deliberate, owned decision with a key and a boundary. Coverage grows where there's a real question to answer, not speculatively.

4. **Bind the process** — attach Lifecycle/Custom-Query semantics so the graph knows *why* entities relate.

5. **Feed Lexicon** — sanctioned relationships and canonical terms flow into BU's semantic layer / knowledge graph, so the discipline compounds across the institution, not just OnBase.

This is **architecture-first** applied to meaning: build the ontology before scaling the analytics on top of it.

7. How you know it's working (governance metrics)

- **Sanctioned-vs-inferred ratio** — share of correlations in answers backed by a sanctioned relationship (target: 100% of *authoritative* claims).
- **Audit pass rate** — % of cross-source claims an auditor can reproduce from the cited rule.

- **Steward coverage** — % of active Normalizers/CrossLinks with a named owner and a confirmation date.
- **False-join catch rate** — candidate joins rejected at the admin-confirm step (a healthy, non-zero number means the guardrail is doing work).
- **Boundary discipline** — % of queries with an explicit term/scope boundary applied.

Confidence, calibration & learning metrics (added per review — operationalized in Document 17):

- **Calibration error** per area — is a confidence band as accurate as it claims?
- **Human-agreement rate** — % of auto/agent decisions a steward upholds.
- **Learned items ratified vs. rejected** — governed corrections entering the ontology.
- **Threshold-by-area & drift alerts** — current autonomy per domain.

8. Summary

The institutional value of AI over OnBase and BU's data is unlocked **only** when the system knows not just the data but **how the business relates that data.** CogniBase encodes that knowledge as a governed ontology — typed entities, sanctioned relationships, vetted keys, bound to process, provenance on every claim, and a steward in the loop. The AI is then free to be brilliant at language and hypothesis, while being **forbidden from inventing the one thing it must never invent: that two records are the same.** That is the difference between an *incredible* analysis and an *auditable* one — and it is the reason CogniBase belongs in an institutional program rather than a demo.

> *See Document 3 (Challenges & Mitigations) for the broader risk register, Document 6 (Knowledge Corpus) for corpus separation, Document 7 (Access Model) for Gates→OPA, and Document 8 (Query Federator) for the join mechanics.*

CogniBase — Challenges & Mitigations

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An honest risk register for bringing OnBase into BU's intelligence stack — and how each risk is resolved.

1. Why this document exists

A credible institutional program names its risks before a sponsor does. This document is the consolidated risk register for CogniBase at BU. Each entry states the **challenge**, its **impact**, the **mitigation** (mapped to a CogniBase mechanism and/or BU's own stack), and the **residual risk** that remains for the pilot to retire. The headline risk — fabricated correlation — is summarized here and treated in depth in Document 2.

2. Risk register

R1 — Fabricated correlation ("convincing but false"; formally *spurious correlation* / *conflation* / *ungrounded inference* — see Document 2 §1a)

- **Impact:** Critical. A fluent, cited, wrong cross-source analysis that a non-expert believes; collapses trust in all AI output when audited.
- **Mitigation:** Governed ontology; deny-by-default joins; entity resolution on vetted keys; inferred-vs-verified labeling; provenance on every claim. **Full treatment in Document 2.**
- **Residual:** Coverage gaps where no ontology edge exists yet — handled honestly by returning *inferred/unverified*, never a guess.

R2 — Entity-resolution ambiguity (no clean shared key)

- **Impact:** High. Sources that *should* join have no reliable common identifier (legacy applicant numbers vs. student IDs; free-text names).
- **Mitigation:** Normalizers declare canonical keys; match-confidence scoring; a record without a sanctioned key is never silently joined — it is surfaced for stewardship or returned inferred-only.
- **Residual:** Manual steward effort to ratify keys for high-value joins; deliberately bounded to where a real question exists.

R3 — OnBase soft/undeclared foreign keys

- **Impact:** High. OnBase enforces relationships in the application layer; a raw schema shows disconnected islands.
- **Mitigation:** Three fused sources — live metadata, Module-Reference-Guide annotations, and `.expk` configuration — with every inferred relationship tagged `confidence` + `source`. (Prior run: 34 MRG-derived + 6,167 inferred soft FKs, each weighted.)
- **Residual:** Inference confidence varies; low-confidence edges are advisory, not authoritative.

R4 — PII leakage across corpora (FERPA)

- **Impact:** Critical. Mixing end-user metadata (possible PII) with reference docs leaks PII into admin answers and blows retrieval relevance.

- **Mitigation: Three-corpus separation** (Operational / Configuration / Solution Documentation); Privacy Transformer (k-anonymity floor, pseudonymization, suppression) before any result reaches LLM context; Gates per group/perimeter. **No HIPAA/PCI ingestion**, by policy.
- **Residual:** Solution docs may contain PII in screenshots → their own hygiene pass.

R5 — Identity, access & OnBase ACLs

- **Impact:** High. Institutional AI must be identity-aware; CogniBase must not let a user see what OnBase would deny them.
- **Mitigation: Entra ID (MSAL PKCE)** SSO; per-user isolation; **OnBase OAuth/AD identity passthrough — CogniBase cannot bypass OnBase ACLs**; audit logging on every access.
- **Residual:** Mapping CogniBase Groups to BU's identity model is a pilot task (Document 13/16).

R6 — Deployment mismatch (local-first vs. EKS/ARM)

- **Impact:** Medium. CogniBase is local-first; BU runs EKS + Graviton (ARM64) + ArgoCD.
- **Mitigation:** Already Docker/compose; publish **ARM64 image + Helm chart**; front with APISIX. Dual packaging keeps the local-first edition (a differentiator) and adds the enterprise edition. (Documents 13.)
- **Residual:** ARM smoke-testing + Helm hardening — scoped, low-risk engineering.

R7 — Open-source licensing (AGPL / SSPL) hygiene

- **Impact:** Medium. Some "open" components (Grafana/Loki = AGPL; MongoDB = SSPL) are unsafe to bundle into a product.
- **Mitigation: Two-edition design.** Bundle only permissive deps; keep AGPL backends **arm's-length/customer-provided** (emit OTLP, BYO dashboard); use **Postgres/JSONB** instead of MongoDB. (Document 13.)
- **Residual:** Counsel review of LICENSE/NOTICE manifests before any commercial release.

R8 — Vendor lock-in vs. "freedom to leave"

- **Impact:** Medium. BU rejects lock-in as a selection criterion.
- **Mitigation:** Plain-file artefacts (schema.json, annotations, anchors as files); open table formats for lake emit; open exit demonstrated in the pilot. Portability is a *feature*, not a concession.
- **Residual:** None material — this is a design strength.

R9 — Hallucinated answers / grounding gaps

- **Impact:** High. The model answers beyond its evidence.
- **Mitigation:** Retrieval-grounded prompts ("use ONLY provided context; cite sources; if insufficient, say so"); source-citation chips; numeric verifier re-executes calculations; refusal on insufficient context.
- **Residual:** Prompt/grounding tuning per corpus — ongoing, measured by audit pass rate.

R10 — Data freshness / staleness

- **Impact:** Medium. Answers grounded on stale anchors mislead.
- **Mitigation:** Hygiene Pipeline with scheduled refresh, health probes (skip + alert on slow OnBase), multi-version anchors with timestamps; freshness surfaced to the user. (Document 9.)
- **Residual:** Cadence tuning per source against OnBase load.

R11 — Scale & performance (2,300+ tables, large corpora)

- **Impact:** Medium. Index build and retrieval latency at institutional scale.
- **Mitigation:** pgvector with SQL filtering; per-corpus collections; caching/rate-limiting (Valkey pattern); incremental updates; nightly publisher to cut source load.
- **Residual:** Capacity sizing during the pilot.

R12 — Neophyte over-trust (the human factor)

- **Impact:** High and often overlooked. The most dangerous failure is a *believable* wrong answer.
- **Mitigation:** UI makes **sanctioned vs inferred visible**; confidence and boundaries shown; "searched N sources" transparency; every claim links to provenance. The product is designed to *protect the non-expert*, not impress them.
- **Residual:** Training/onboarding for end users; an explicit "how to read an answer" guide.

R13 — Coordination & credentials with BU's REST POC team

- **Impact:** Medium. Two teams on the same Hyland API; no contact/credentials = stall.
- **Mitigation:** "Coordinate, don't integrate" model; one shared IT credential request; clear ownership split (BU owns DocType inventory + credentials; CogniBase owns the OnBase domain layer). (Document 16.)
- **Residual:** Depends on BU naming a sponsor and issuing credentials — the gating signal.

R14 — LLM cost governance

- **Impact:** Low–Medium. Uncontrolled model spend.
- **Mitigation:** Curated heterogeneity — route cheap/local models for routine work, frontier only when needed; token/latency metrics per conversation; cost transparency. (Document 12.)
- **Residual:** Budget thresholds set with BU.

R15 — Miscalibrated confidence & feedback poisoning

- **Impact:** High. Using the model's self-confidence as a review gate (it is uncalibrated), or letting user corrections auto-learn, can teach the system falsehoods at scale.
- **Mitigation:** Evidence-based confidence (not LLM self-confidence); corrections are **steward-ratified proposals**, not auto-absorbed; feedback authority tracks stewardship; thresholds change only on measured calibration and auto-tighten on drift. **Full treatment in Document 17.**
- **Residual:** Calibration must be measured per area before autonomy is raised.

3. Risk posture summary

Severity	Risks	Net position
Critical	R1 (false correlation), R4 (FERPA PII)	Addressed by the trust layer (Doc 2) + corpus separation/Privacy Transformer — the core of the design
High	R2, R3, R5, R9, R12, R15	Mechanisms exist; pilot retires the residuals
Medium	R6, R7, R8, R10, R11, R13, R14	Engineering + coordination, low conceptual risk

The pattern is deliberate: **the two Critical risks are exactly the two CogniBase was architected to solve** (governed correlation and corpus/PII separation). The remaining risks are the ordinary work of taking a capable system to institutional grade — which is what the pilot is for.

4. What would make us pause

Honest engagement means naming the conditions under which CogniBase should *not* proceed: no named data steward (the ontology needs an owner); a mandate to ingest HIPAA/PCI into the lake (out of policy); or a requirement to assert cross-source relationships without sanctioned keys (the one thing the design refuses to do). Each of these is a conversation, not a workaround.

> *Cross-references: Document 2 (ontology), Document 6 (corpus), Document 7 (access/Gates→OPA), Document 8 (federation), Document 10 (provenance/audit/FERPA), Document 13 (deployment/licensing), Document 16 (coordination/pilot).*

CogniBase — Architecture & Atlas Alignment

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1. Purpose

CogniBase is a local-first, vendor-neutral system that lets BU staff (1) **see** the full OnBase schema without SQL access, (2) **search and retrieve** OnBase documents through a familiar UI without an OnBase client, (3) **ask** natural-language questions across schema, metadata, and document text, and (4) **report** on dashboards, lifecycles, and document populations. This document shows its internal architecture and how each part lands as a native citizen of **Atlas**.

2. The layer cake

```
┌── Channels ─────────────────────────────────────────────┐
│  Retrieval UI · Schema browser · Reports · Assist chat    │
├── Agent / Orchestration ────────────────────────────────┤
│  RAG query engine · Query Federator · Numeric verifier    │
├── Knowledge / Trust ────────────────────────────────────┤
│  Ontology (Normalizers·CrossLinks) · 3-corpus index       │
├── Access / Governance ──────────────────────────────────┤
│  5-layer access model · Gates(→OPA) · Privacy Transformer │
├── Sources ──────────────────────────────────────────────┤
│  OnBase (Unity/REST/.expk) · SQL · institutional APIs     │
└── Provenance & Audit (cross-cutting, L1–L4) ────────────┘
```

Every answer flows **bottom-up through the trust layer** before it reaches a channel — never source-to-channel directly. This mirrors BU's intelligence-stack rule (*agents never reach raw systems*).

3. Atlas alignment

CogniBase layer	Atlas home	Mechanism
Agent / RAG / Federator	**Nexus**	Registered MCP tools; Nexus/Regent agents call them
Ontology / knowledge	**Lexicon**	Normalizers + sanctioned relationships feed BU's knowledge graph
3-corpus index	**Cortex** (+ local)	Curated entities emit to Iceberg/pgvector; Trino-queryable
Access / Gates	**Forge / OPA**	Gates expressed as OPA policy; deny-by-default
Channels / API	**Lattice / APISIX**	FastAPI behind APISIX; Entra ID auth
Audit	**Forge audit sinks**	L1–L4 trail ships to BU's Vector/Loki sinks

4. Two-VPC fit

BU separates a **Data Engineering VPC** (lake, Trino, OPA) from an **AI Engineering VPC** (Nexus, MCP, pgvector), joined only through APISIX over Nitro-encrypted peering. CogniBase fits both faces:

- **AI-plane (lead):** an MCP server in the `mcp-gateway-registry` — agents call CogniBase tools; lightest, native.

- **Data-plane (Phase 2):** an "OnBase API" behind APISIX, peer of Person/Course API, emitting curated entities to Cortex.

(Full integration mechanics in Document 11; deployment in Document 13.)

5. Deployment shape

Single-process FastAPI + uvicorn; ChromaDB→**pgvector** for the index; vendor-pluggable LLM router; plain-file artefacts per profile (`schema.json`, annotations, anchors, audit folders). Local-first for the desktop edition; container + Helm (ARM64) for the enterprise/BU edition — **one codebase, edition profiles** (Document 13).

6. Sync vs. cache strategy

OnBase is the system of record; CogniBase holds **derived, governed** representations. Configuration (`.expk`, schema) is **synced** on change; operational data is **anchored** through the hygiene pipeline with timestamps and multi-version retention (Document 9); document binaries are fetched **on demand** and never cached beyond policy. Freshness is surfaced to the user — staleness is shown, never hidden.

> *CogniBase's architecture is deliberately **Atlas-shaped**: a governed knowledge layer over OnBase that plugs into Nexus, Cortex, Lexicon, and Forge through their own interfaces, not around them.*

CogniBase — Components Overview

Document 5 of 18 · BU-Aligned Library v2 · Renne Santiago

1. What CogniBase delivers

Unbound but anchored. Cross-departmental analysis OnBase cannot do natively — with every claim citing its source and every cross-source correlation backed by a sanctioned relationship (Document 2). The component set below exists to make that promise operational and auditable.

2. Component catalog

Layer	Component	Purpose	Status
Channels	Retrieval UI	OnBase-Web-Client-like document search/preview, no client install	Live
	Schema browser	Navigable HTML of tables/columns/keys/relationships	Live
	Reports	`.docx`-templated + ad-hoc RAG-driven generation	Designed (M1x)
	Assist chat	Contextual NL help on schema and retrieval	Live
Agent	RAG query engine	Retrieve → synthesize with citations	Live
	Query Federator	Decompose → sanctioned sub-queries → join on Normalizers	Designed (Doc 8)
	Numeric verifier	Re-executes every numeric claim before it ships	Designed (M32)
Knowledge	Ontology (Normalizers/CrossLinks)	Semantic equivalence + vetted joins	Core
	3-corpus indexer	Operational / Configuration / Solution Documentation	Core
Access	5-layer access model	Constraint→Normalizer→CrossLink→Gate→Privacy Transformer	Core
	Privacy Transformer	k-anonymity, masking, suppression before egress	Core
Sources	OnBase clients	Unity API + REST + `.expk` config parser	Live/partial
	Metadata collector	DocTypes, KeywordTypes, Lifecycles, dashboards	Live (M9)
Platform	LLM vendor router	Pluggable Claude/GPT/Gemini/local	Live
	MCP server	Tool surface for Nexus/Regent	Proposed (Doc 11)
	Scheduler	APScheduler-driven hygiene/refresh jobs	Stub→M34
Cross-cutting	Provenance & audit	L1–L4 trail on every record/report	Core (Doc 10)

Review & learning components

- **Review queue** — confidence-gated items awaiting steward ratification.
- **Feedback & learning loop** — governed capture of corrections into sanctioned items (Document 17).
- **Review agents** — Inspector (consistency/policy) and Librarian (provenance/placement) for triage at scale.

3. Discovery toolkit (shared with MapSnap)

The schema-extraction, soft-FK inference, and enrichment pipeline run natively in **both CogniBase and MapSnap** — the same engine-agnostic discovery layer, pointed at OnBase here and at any enterprise database in MapSnap. This is the product-family backbone and a credibility signal: the OnBase domain layer is built on a proven, reused foundation.

4. What is BU-new vs. carried forward

- **Carried forward (proven):** schema map, soft-FK inference, three-corpus RAG, vendor router, Privacy Transformer, audit trail.
- **BU-aligned additions (this library):** MCP tool server (Nexus-native), pgvector index, Entra ID identity, OPA-expressed Gates, lake/Iceberg emit, observability (OTLP). See Documents 11–14.

> *The catalog is intentionally small and sharp. CogniBase is not a platform trying to be everything — it is the **governed OnBase knowledge layer**, and every component earns its place against that single job.*

CogniBase — Knowledge Corpus: The Three-Corpus Model

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1. Why a corpus design at all

Retrieval quality and data governance are the same problem viewed from two sides. If everything is indexed into one vector space, PII leaks into admin answers, refresh cadences collide, trust boundaries blur, and relevance degrades. CogniBase therefore separates knowledge into **three corpora with one shared engine** — each with its own audience, cadence, hygiene, and gate posture. This is the substrate the ontology (Document 2) reasons over.

2. The three corpora

Corpus	Question it answers	Examples	Audience	Index
Operational Data	"What is in the system right now?"	OnBase document instances, keyword values, lifecycle states (via REST)	End user (gate-filtered)	`cognibase_main`
OnBase Configuration	"How is the system structured?"	`.expk` packages, `schema.json`, Hyland manuals	Admin / DBA	`cognibase_config`
Solution Documentation	"Why does it look like this and how should it behave?"	BU design docs, SOPs, Excel mappings, training decks	Mixed (gate-controlled)	`doc_corpus`

3. Why three and not one

Putting end-user metadata (possible PII) into the same space as static reference docs would (a) leak PII into admin chat, (b) wreck retrieval relevance, (c) couple unrelated refresh cadences, and (d) collapse trust boundaries — configuration is admin-uploaded, operational data passes through hygiene, solution docs may carry PII in screenshots. Three logical corpora, one shared retriever, **gates applied at query time.**

4. Solution Documentation — the institutional knowledge layer

This corpus is where BU's *business meaning* lives: narratives, SOPs, mapping sheets, training materials. Materialized as doc_corpus anchors with **bindings to OnBase entities**, gate-pinned, and feeding **business-rule verification** — the bridge between "the data" and "how the business relates the data," which is exactly the gap that causes fabricated correlation (Document 2). An admin defines a watched folder per anchor; dropped files auto-ingest through the hygiene pipeline (Document 9).

5. Cross-corpus retrieval

When the agent retrieves from all three simultaneously, results are ranked with **corpus-aware weighting** and **gate filtering first** (deny-by-default): operational hits are PII-masked per policy, configuration is admin-only, solution docs cite down to paragraph/cell. Every retrieved chunk carries its corpus, source, and gate decision into the provenance trail.

6. Alignment to BU

- **Cortex / lake:** curated Operational anchors can emit to Iceberg/pgvector for Trino-queryable analytics.
- **Lexicon:** Solution Documentation bindings and canonical terms feed BU's semantic layer / knowledge graph.
- **FERPA:** corpus separation + Privacy Transformer is the structural guarantee that regulated content cannot cross into an unprivileged answer.

> *The three-corpus model is what lets CogniBase be simultaneously **useful to end users**, **safe under FERPA**, and **truthful across sources** — without those goals fighting each other.*

CogniBase — Access Model: Deny-by-Default & Gates → OPA

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1. Principle

Every layer adds intent, and **deny-by-default** is the rule whenever a layer is silent. The model is stacked so that "not explicitly allowed" always means "denied" — the posture BU enforces platform-wide with OPA.

```
Layer 5 — Privacy Transformer     (what the output may look like)
Layer 4 — Cross-Reference Gate    (who, where, how — policy)
Layer 3 — CrossLink               (which DocTypes may join — admin intent)
Layer 2 — Normalizer              (which keywords are the same — equivalence)
Layer 1 — DocType Constraint      (respect the .expk grouping — default)
```

2. The five layers

1. **DocType Constraint** — on `.expk` import, OnBase's grouping is honored as-is; no cross-cuts unless authorized above.

2. **Normalizer** — declares semantic equivalence (`AP_Name ≡ HR_Name ≡ OAD_Name → canonical:PersonName`); auto-discovered, **admin-confirmed**, or Excel-uploaded. This is where "are these the same entity?" is answered by a vetted rule, not a guess (Document 2).

3. **CrossLink** — admin-vetted joins between DocTypes (CogniBase's Normalizer-aware Custom Query); may reuse existing OnBase Custom Queries by ID.

4. **Cross-Reference Gate** — per CogniBase Group, per perimeter (User vs AI), declaring `source` (lake / live_rest / hybrid / deny) and `transform` (raw / anonymize / aggregate_only / deny), with field-level overrides.

5. **Privacy Transformer** — k-anonymity (k≥5 floor), pseudonymization, generalization, suppression on results **before** they reach LLM context or the user.

3. Mapping to BU: Gates → OPA, Groups → Entra

- **Gates as OPA policy.** Each Gate is expressible as a declarative Rego policy, so access decisions are externally auditable and enforced by BU's own engine rather than bespoke code — policy decoupled from application.
- **CogniBase Groups ↔ Entra ID.** Groups are decoupled from AD and bound to **Entra ID** identities (MSAL PKCE), giving per-user isolation and identity-aware access. CogniBase **cannot bypass OnBase ACLs** — OnBase OAuth/identity passthrough is authoritative.
- **Audit.** Every layer decision (allow/deny/transform) is written to the L1–L4 trail (Document 10) and can ship to BU's audit sinks.

Feedback authority & learned items

Corrections enter the access model as **governed proposals**: anyone with access to an area's data may **flag**, but only a designated **domain steward** may **ratify** a correction into a sanctioned Normalizer/CrossLink. Ratified items carry provenance and re-enter via the Hygiene Pipeline (Document 9). Authority tracks **stewardship, not mere data access** — the operational rule behind the learning loop (Document 17).

4. Why this matters for trust

The access model is not only a security control — it is half of the **anti-fabrication** design. Layers 2–3 decide *whether two things may be related at all*; Layers 4–5 decide *what may be shown*. Together they ensure the agent can neither invent a relationship nor over-disclose one. Security and truthfulness are enforced by the same stack.

> *Deny-by-default is the quiet backbone of institutional trust: the system's default answer to "can these connect / can this be shown?" is* ***no****, until a named owner says yes.*

CogniBase — Query Federator: Sanctioned Cross-Source Joins

Document 8 of 18 · BU-Aligned Library v2 · Renne Santiago

1. The problem the Federator solves

OnBase Custom Queries cannot join across keyword types that BU has aliased with prefixes (`AP_Name` vs `HR_Name` vs `OAD_Name`). More deeply: an AI must never join sources on *similarity*. The Query Federator is the engine that makes every cross-source join **sanctioned, keyed, and auditable** — it is Document 2's discipline turned into execution.

2. Pipeline

```
Agent intent
  → Logical Query (canonical entity names)
  → Logical Query Planner (resolve Normalizers · find CrossLink · check Gate)
  → Federation Plan (N sub-queries + a join recipe)
  → Per-source dispatch (lake anchor / live REST / hybrid)
  → Join inside CogniBase on the Normalizer key (never on name/embedding)
  → Privacy Transformer → cited result
```

The planner's first act is **authorization**: if no CrossLink sanctions the join, or a Gate denies it, the plan **fails closed** — the agent returns an *inferred/unverified* note, not a fabricated join.

3. Federation Plan — worked shape

For "students with stuck FA documents and their enrollment status":

1. Sub-query A — OnBase FA DocType where Lifecycle = "In Review" > 7 days → returns `canonical:StudentID` set.

2. Sub-query B — SIS enrollment for those StudentIDs (current term boundary).

3. Join on `canonical:StudentID` only (CrossLink #N), Gate-checked, term-bounded.

4. Result cites the CrossLink, the Lifecycle state, and the boundary.

4. Reusing existing OnBase Custom Queries

Where OnBase already encodes a join, the Federator **issues that Custom Query by ID** (one round-trip — OnBase did the join) rather than re-implementing it. This respects BU's existing investment and avoids duplicate logic.

5. Open design choices (for BU discussion)

- **live_rest ⋈ lake** in one plan — allowed, with freshness disclosed per source.
- **Plan materialization** — repeated plans can be cached as reusable, versioned artefacts (with the same Gate re-evaluated on each run).

- **Trino as substrate** — for the SQL-source portion, BU's Trino could execute federated sub-queries; CogniBase contributes the Normalizer-aware join recipe and the OnBase domain semantics Trino lacks. A natural division of labor with Cortex.

6. Why it belongs at BU

The Federator is the concrete answer to "how do we get cross-departmental insight from OnBase + institutional data **without** the fluent-but-false analysis?" Every join it performs is one a named admin authorized, on a key the business actually uses, within a stated boundary, recorded for audit. It is the operational heart of governed correlation.

> *The Federator's golden rule:* ***it would rather return less than assert a join no one sanctioned.***

CogniBase — Hygiene Pipeline & Scheduled Jobs

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1. What a hygiene job does

A scheduled job:

1. **Pulls** targeted data from OnBase via REST (one or more sub-queries — possibly federated).

2. **Runs** it through a configurable **Hygiene Pipeline** of steps.

3. **Materializes** the result as a versioned, gate-pinned **anchor**.

4. **Records** everything to the audit trail and run log.

Anchors — not live queries — are what the agent reasons over for operational data, so answers are reproducible and policy-bounded.

2. Why per-job hygiene + multi-version anchors

Different data needs different treatment, and the same source may appear at different transformation levels (raw / hybrid / fully anonymized) pinned to different Gates. Multi-version anchors let an admin question see masked data while an aggregate dashboard sees suppressed data — from one source, governed differently, each version timestamped.

3. Hygiene step library (representative)

Step	Purpose
Normalize keys	Map source keywords to canonical entities (Normalizers)
PII detect + mask	Find and transform regulated fields before indexing
Deduplicate	Collapse repeated records on the sanctioned key
Validate / integrity-check	Reject malformed or out-of-contract rows
Embed	Vectorize for retrieval (pgvector)
Profile	Sample + classify for the value catalog

Steps are admin-configurable per job; heavy steps run in the background, fast steps inline.

4. Scheduling & safety

- **Cadence** per job (cron via the scheduler), tuned to source volatility.
- **Health probe** → `GET /onbase/.../healthcheck`; if p95 latency exceeds threshold, **skip with a recorded reason and alert the admin** — never hammer a stressed OnBase.
- **Sequential job execution** (parallelism is *within* a job via `max_concurrent_rest`, not across jobs) to bound OnBase load; a nightly publisher cuts source load by ~90%.
- **Per-step timeout** to catch a hanging step (e.g., a flaky embed vendor).
- Hygiene runs on **every ingestion path**, not just scheduled — drag-drop imports get the same treatment.

Learned items re-enter through hygiene

Steward-ratified corrections (Document 17) re-enter as versioned anchors through this same pipeline — a learned item is just a new sanctioned input with provenance, subject to the same hygiene, gating, and audit. Nothing is “learned” outside the governed path.

5. Alignment to BU

- **Dagster fit.** Each hygiene job maps cleanly onto a Dagster asset (BU's orchestrator in Cortex) — observable, retryable, lineage-tracked — should BU prefer to run pipelines on its own platform.
- **Freshness contract.** Anchor timestamps surface to the user; staleness is shown.
- **Audit.** Every run, skip, and transform is in the L1–L4 trail (Document 10).

> *The hygiene pipeline is what keeps governed data **fresh without becoming unsafe** — the operational complement to the trust layer.*

CogniBase — Provenance, Verification, Audit & FERPA

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1. The promise

Every number CogniBase reports can answer three questions: **Where did this come from? How was it computed? Who was allowed to see it?** An auditor must be able to reconstruct any claim without the original author present. This is the difference between an analysis BU can act on and one it merely admires.

2. The mechanisms

Mechanism	What it does
Provenance tagging	Every record carries "where did this number come from" — source, anchor, version, retrieval id
Sanctioned-join citation	Every cross-source claim cites the CrossLink/Normalizer that authorized it (Document 2/8)
Numeric verifier	Re-executes every numeric claim before it ships; mismatch → flag, not publish
Audit trail	Every report ships with an audit folder an auditor can replay
Gate decision log	Every allow/deny/transform recorded with the policy that decided it

3. Audit levels (L1–L4)

Level	Scope	Example
L1	The answer	Final text + citations
L2	The retrieval	Which chunks/anchors, which scores
L3	The computation	Sub-queries, joins, the verifier's recomputation
L4	The policy	Gate decisions, identity, transforms applied

An auditor descends from L1 to L4 to fully reconstruct *why the system said what it said.*

4. Audit folder layout

Each report writes a self-contained folder: the answer, the cited sources, the federation plan, the verifier's recomputation, and the gate/identity record — a portable, plain-file evidence package (no lock-in).

5. FERPA & data classification (folded in)

- **No HIPAA/PCI ingestion**, by policy.

- **FERPA-classified access is audit-trailed**; transcript/education-record *content* is gated — existence/counts may be returned, content only under explicitly raised policy with a recorded agreement.
- **Privacy Transformer** (k-anonymity floor, masking, suppression) runs before any result reaches the model or user.
- Identity is **Entra ID**; CogniBase honors OnBase ACLs (cannot over-disclose).

Provenance of corrections & confidence calibration

Every learned item records **who proposed it, who ratified it, when, and on what evidence** — corrections are first-class audited events. Audit also tracks **confidence calibration per area** (does a band mean what it claims?) and **threshold-change events**, so raising an area's autonomy is itself an auditable, reversible decision (Document 17).

6. Open choices (for BU)

- **Tolerance bands** for numeric verification — exact / ±0.01 / ±1% tiers vs. BU's audit context.
- **Retention** — how long audit folders live (tied to report / compliance period / indefinite).
- **Cross-report audit search** — "show every 2026 report that used anchor X" (future).
- **Audit-sink integration** — ship L1–L4 events to BU's **Vector → Loki/Mimir/Tempo** stack via OTLP for institution-wide observability.

> *Provenance is not paperwork — it is the mechanism that makes a believable answer also a verifiable one, which is the whole point at an institution under FERPA.*

CogniBase — Platform Integration: MCP & Nexus

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1. The change from the original design

The original library described integration via a **LibreChat gateway**. BU's platform has since standardized on **MCP (Model Context Protocol) as the architectural spine** — Nexus exposes an `mcp-gateway-registry` and a `Nexus MCP Control Plane` that route agent tool-calls. This document supersedes the LibreChat approach: **CogniBase integrates as an MCP server.** (LibreChat/OpenAI-compatible chat remains available for *local desktop* use only — Document 12.)

2. The integration, concretely

CogniBase exposes its capabilities as **MCP tools** and **registers in BU's `mcp-gateway-registry`**. A Nexus or Regent agent then calls them like any other Atlas tool — no bespoke connector:

MCP tool	Backed by	Returns
`cognibase.search`	RAG retriever	gate-filtered passages + citations
`cognibase.ask`	RAG query engine	synthesized, cited answer
`cognibase.schema_explain`	config corpus	plain-English schema/DocType explanation
`cognibase.federated_query`	Query Federator	sanctioned cross-source result (Document 8)
`cognibase.list_doctypes`	schema/.expk	DocType inventory

A runnable scaffold exists (`C:\CogniBase\proposals\mcp_server.py`) wrapping the live retriever/query-engine/router.

3. Where it runs (two-VPC)

- **AI Engineering VPC (lead):** the MCP server sits beside Nexus; the Control Plane routes to it; identity via **Entra ID**, transport behind **APISIX**, policy via **OPA**.
- **Data Engineering VPC (Phase 2):** optionally also expose an "OnBase API" behind APISIX as a peer of Person/Course API, for data-plane analytics and lake emit.

4. Why MCP-first is the right call

- **Native:** it is the exact interface BU's agents already speak — zero integration tax.
- **Governed:** tool calls inherit APISIX auth, OPA policy, and the audit trail.
- **Reversible:** an MCP tool is a thin, replaceable surface — it honors "freedom to leave."
- **Portable (Plan B):** the same MCP server works for any MCP-capable host, so the integration investment survives even if BU does not.

5. Streaming & UX parity

CogniBase supports token-by-token streaming, source-citation chips, and "searched N sources" transparency — matching the Nexus channel experience, so a CogniBase-backed assistant feels native inside BU's widget/Teams/Slack surfaces.

> *Integration is not a bridge bolted onto BU's platform — it is CogniBase **speaking BU's own protocol**, registered in BU's own registry, governed by BU's own policy engine.*

CogniBase — Model Strategy & Curated Heterogeneity

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1. Principle: no single bet

CogniBase shares BU's stated commitment to **curated heterogeneity** — multiple models, chosen per task on fit, cost, capability, and freedom to leave. The LLM is a **pluggable component behind a vendor router**, never hardcoded. Swapping or adding a provider is a config change.

2. The provider interface

A single `VendorRouter` fronts concrete adapters with a uniform `chat()` / `embed()` contract:

Adapter	Use
`claude`	Frontier reasoning, synthesis
`openai`	Alternate frontier / embeddings
`gemini`	Alternate frontier
`ollama`	Local, regulated, zero-egress
`openai_compatible`	LM Studio / llama.cpp / vLLM / any OpenAI-compatible server

Adapters are checkbox-active and multi-active; keys live outside code (Document 13).

3. Mapping to BU's providers

BU runs models through **AWS Bedrock** (Claude) and **Azure OpenAI** (GPT), with Gemini planned. CogniBase's router targets the same providers through their endpoints, so model governance stays in BU's hands:

BU path	CogniBase adapter
Claude via Bedrock (IAM)	`claude` / `openai_compatible` to the Bedrock endpoint
GPT via Azure OpenAI (key)	`openai` to the Azure deployment
Local / regulated	`ollama` / `openai_compatible`

4. Routing economics (efficiency by design)

Not every task needs a frontier model. CogniBase routes **cheap/local models for routine work** (schema explanation, classification, draft retrieval) and **frontier models only where reasoning demands it** (cross-source synthesis, ambiguous questions). Per-conversation **token + latency metrics** make cost transparent and tunable — aligning with BU's per-school cost-transparency goal.

5. Why this matters for trust

Model heterogeneity also reduces single-model blind spots: a sanctioned correlation can be **cross-checked across providers**, and the numeric verifier (Document 10) re-executes calculations independent of any one model's confidence. Diversity is a correctness tool, not just a procurement stance.

> *The model is the most replaceable part of CogniBase by design. The durable value is the* ***ontology, governance, and provenance*** *around it — which is exactly where BU wants its institutional investment to sit.*

CogniBase — Deployment, Operations & Licensing

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1. Two editions, one codebase

CogniBase ships as **edition profiles of a single codebase**, not two forks. Edition-variable seams (vector store, identity, telemetry, policy, gateway) use the same pluggable-adapter pattern as the LLM router:

Edition	Target	Vector	Identity	Edge	Observability	Deploy
Enterprise (Atlas)	BU / institutions	pgvector	Entra ID (OIDC)	APISIX + OPA	OTLP → customer Grafana	Helm / EKS / ARM64
Neutral / Core	commercial or free	pgvector/SQLite	OIDC/local	optional	OTLP only (BYO dashboard)	Desktop / Docker

2. BU-grade hardening (the commercial-grade uplift)

Adopting BU's own permissively-licensed stack moves CogniBase from local POC to institutional product:

- **pgvector** (replaces ChromaDB) — ACID, concurrency, SQL filtering, backups; matches BU's Person/Course indices.
- **OpenTelemetry + Prometheus** instrumentation → traces/metrics/structured logs.
- **OIDC / Entra ID** (`authlib` already a dependency) — enterprise SSO.
- **APISIX** gateway (TLS, rate-limit, API-key consumers) + **OPA** for Gates.
- **External Secrets** — keys out of `settings.json`.
- **Helm + ARM64** image for Forge/Graviton.

3. Licensing hygiene (open-source done safely)

CogniBase follows BU's own rule — **permissive only, no copyleft bundled into the product**:

- **Bundle freely (permissive):** pgvector/Postgres, OpenTelemetry SDK, Prometheus client, APISIX, OPA, authlib, Valkey, Dagster/Prefect, Iceberg/Trino.
- **Arm's-length / customer-provided (AGPL):** Grafana/Loki/Tempo/Mimir — emit OTLP, BYO dashboard; never vendored. (Same separate-process discipline AutoPDF uses for GPL engines.)
- **Avoid (SSPL):** MongoDB → use **Postgres/JSONB** (which pgvector already brings in).

> *The two-edition design lets BU's edition use the full open stack while the public edition stays legally headache-free — with no GPL/AGPL/SSPL in the shipped artefact.*

4. Operations

- **GitOps** deploy (ArgoCD-compatible Kustomize/Helm); rollback via revision history.
- **Health probes** (liveness/readiness), HPA autoscaling, per-conversation metrics.
- **Secrets** via External Secrets Operator / Vault; no embedded credentials.

- **Backups** — pgvector + plain-file profile artefacts; portable by construction (freedom to leave).

> *Licensing note: counsel should confirm the LICENSE/NOTICE manifests before any commercial release. The internal contingency plan governs which deep IP is exposed.*

CogniBase — API & Tool Contract: REST + MCP

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1. Two faces, one engine

CogniBase exposes the same governed engine through two contracts: a **REST API** (for the UI, reports, and data-plane integration) and an **MCP tool surface** (for Nexus/Regent agents). Both pass through the access model and audit trail — there is no ungoverned back door.

2. REST surface (representative)

Method · Path	Purpose
`POST /schema/extract`	Extract schema from a live connection
`POST /metadata/import-expk`	Ingest OnBase `.expk` configuration
`POST /query`	RAG answer (grounded, cited)
`POST /federate`	Run a Federation Plan (sanctioned join)
`GET /anchors` / `POST /anchors/refresh`	Inspect / refresh hygiene anchors
`POST /reports/generate`	Templated or ad-hoc report + audit folder
`GET /health` · `GET /version` · `GET /info`	Ops/JSON (standard health endpoints)

Responses stream via **SSE** (`content_delta`, `tool_call_start`, `tool_result`, `auth_required`, `done`) — matching Nexus's event contract.

3. MCP tool surface

`cognibase.search` · `cognibase.ask` · `cognibase.schema_explain` · `cognibase.federated_query` · `cognibase.list_doctypes` (Document 11). Each tool description is admin-managed and visibility-gated; the same Gate/Privacy-Transformer pipeline runs before any tool result returns.

4. Authentication & config

- **Auth:** Entra ID (MSAL PKCE) for users; APISIX consumer keys for widget/agent deployments; OnBase OAuth passthrough for source access.
- **Config API:** ETag-cached config (agents/deployments/models/tool-endpoints) the agent backend polls — no restart on change, mirroring Nexus's admin-config pattern.
- **Smart auth:** a synthetic tool lets the model request identity only when a question needs it.

5. Contract stability

The OnBase REST shapes are pinned to API version 1; field anomalies are documented in coordination with BU's REST POC team (Document 16). The MCP contract follows the published MCP spec, so BU's gateway treats CogniBase like any registered tool server.

> *The contract is deliberately boring and standard — REST + SSE + MCP — because predictability is what makes a component safe to adopt and safe to leave.*

CogniBase — Beyond OnBase: Lake, Semantic Layer & Knowledge Graph

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1. The mission shift

OnBase is rigid: walled tables, hard to combine with other sources, hard to find patterns across the full content. The strategic ambition is to **exceed OnBase** — lift what it exposes into a governed vector + lake layer where AI can correlate, infer, and report across boundaries OnBase itself cannot cross. Crucially, this expansion **inherits the trust layer** (Document 2): more reach, not more fabrication.

2. Four additions (governed, not speculative)

1. **Lake emit** — curated OnBase entities written to **Iceberg-on-S3** (Cortex), Trino-queryable, lineage-tracked. Open table format = no lock-in.

2. **Entity layer** — canonical entities (Person, Student, Vendor, Asset…) resolved on sanctioned keys, the nodes of a knowledge graph.

3. **Semantic layer** — shared business definitions feeding BU's **Lexicon** (Cube/Superset/OpenMetadata), so "active enrollment" means one thing everywhere.

4. **Agentic reach** — Regent-style headless workflows over the governed graph (freshness monitoring, anomaly triage), each tool call gated and audited.

3. Alignment to BU's roadmap

BU has published an **"Ontology Investigation"** and a 12-month goal of an **enterprise knowledge graph linking metrics, data, and policy.** CogniBase's Normalizers/CrossLinks are knowledge-graph edges in waiting; its OnBase domain modeling is the hardest, most strategic input to that graph — the layer BU has self-identified as weakest. CogniBase contributes OnBase's slice of the institutional ontology, evidence-first and steward-confirmed.

4. Local LLM strategy (portability)

"Beyond OnBase" does not mean "cloud-only." The `openai_compatible` adapter speaks the de-facto local-server protocol (LM Studio / llama.cpp / vLLM / Ollama-compat), so regulated correlation can run entirely on-prem when policy requires — the same engine, different backend.

5. The `.expk` + soft-FK foundation

A system without enforced foreign keys keeps its semantics in code, conventions, and configuration. Once CogniBase extracts those (`.expk` intent + MRG annotations + inference) into a vector + graph layer, **the relationships become first-class for the AI** — OnBase cannot natively ask "all docs related to Vendor X across all DocTypes and lifecycles," but the governed graph can.

6. Sequencing (Plan-A/Plan-B aware)

Build the **portable core** (entity layer, semantic feeds, lake emit on open formats) first — it serves BU *and* every other OnBase customer. Keep BU-specific wiring thin and late. The "beyond OnBase" vision is therefore also the diversification hedge: the more general the knowledge layer, the less any single customer gates its value.

Across BU's systems of record. The lift-into-a-lake mission naturally reaches BU's wider federation — MyBU/Campus Solutions, SAP, the facilities cluster (CAMMS/PMWeb/25Live/FMS), Blackbaud, Kuali/Huron — but only through sanctioned relationships. **Document 18** maps that landscape and the inclusion test (natural / policy-bounded / excluded) that governs which systems CogniBase may touch.

> *Beyond OnBase is not scope creep — it is the same governed discipline, extended from one content store to the institution's connected knowledge.*

CogniBase — Coordination & Pilot Plan

Document 16 of 18 · BU-Aligned Library v2 · Renne Santiago

1. Working model with BU's OnBase REST POC team

BU has a team exploring the OnBase REST API with strong REST/HTTP skills but limited OnBase domain depth. CogniBase brings the domain layer (DocType modeling, Custom Query semantics, Lifecycle/Workflow awareness, Normalizers, Gates, Hygiene) on top of plumbing both teams hit. The relationship is **coordinate, not integrate** — same upstream Hyland API, shared notes, no duplicated work.

Concern	Owner
Hyland identity/tenant config, credentials	BU IT
DocType / KeywordType departmental inventory	BU REST POC team
OnBase domain modeling, Normalizers, Gates, Federator, hygiene, RAG, audit	**CogniBase**
Document Management REST client	Either (shared/vendored)

2. What we ask of BU (the gating signals)

1. A **sponsoring team** and a **named technical contact**.

2. **One IT credential request** for a shared **non-production** OnBase instance usable by both teams (draft ready).

3. A **named data steward** to ratify ontology relationships (the ontology needs an owner — Document 2).

3. Pilot scope (3–4 weeks)

A narrow, high-value use case (e.g., FA-document lifecycle vs. SIS enrollment) demonstrating, in BU's terms:

1. **Governed correlation** — a real cross-source question answered through sanctioned relationships, inferred-vs-verified visible.

2. **Auditability** — every claim reconstructable L1–L4 (FERPA-grade).

3. **Atlas-native integration** — CogniBase reachable as an MCP tool from a Nexus agent.

4. **No lock-in** — plain-file portability demonstrated.

4. Critical path

Step	Gate
Named sponsor + steward	BU
Credentials + non-prod instance	BU IT
MCP server registered in `mcp-gateway-registry`	CogniBase (scaffold exists)
pgvector + Entra ID identity wired (BU mode)	CogniBase

Seed ontology from .expk + Person CDM; steward-confirm Normalizers	Joint
Pilot use case live + audit demo	Joint

5. Success criteria

- 100% of *authoritative* cross-source claims backed by a sanctioned relationship.
- Auditor reconstructs any claim from the cited rule.
- A Nexus agent answers an OnBase question via CogniBase's MCP tool.
- BU confirms the open-exit (portability) test.

6. Roadmap beyond pilot

GA OnBase coverage → lake/Iceberg emit into Cortex → semantic feeds into Lexicon → Regent-style agentic workflows. Sequenced **portable-core-first** so value compounds regardless of any single milestone (internal contingency plan governs pace and IP exposure).

Roadmap fit. CogniBase advances items already on BU's funded 2026 roadmap rather than adding a new one — **Appendix B** crosswalks it to all four lanes (strongest in AI Engineering, concrete in Data Engineering and Enterprise Architecture, supporting in Essential Services).

7. Co-Development Model — a governed tandem build

BU's stated posture is *"AI has changed build vs. buy — we build in-house."* CogniBase takes that seriously: the strongest engagement is **not a product sale but a governed co-development** — BU builds *with* the CogniBase team, keeping ownership and skills, on the design-partnership pattern BU already runs. Appendix B shows the shared roadmap items; this section is how the two sides build them together.

The IP boundary (what makes co-development clean):

Co-built & BU-owned	Licensed CogniBase core
The BU ontology content — which DocTypes/keywords/entities relate, the canonical-key mappings, stewardship decisions	The governed-correlation engine — Normalizers/CrossLinks/Gates, Query Federator, Privacy Transformer, hygiene + discovery pipeline
Use-case adapters, integration wiring, the BU knowledge graph	The reusable, maintained, evolving platform (also powers MapSnap)

The seam is the edition/adapter boundary already in Document 13: **BU owns the institution-specific knowledge; the engine stays a licensed, maintained core** — good for BU too, who get an evolving engine rather than a fork to maintain alone.

In practice:

- **Forward-deployed.** The CogniBase team embeds with BU's OnBase REST POC team (plumbing) and the domain stewards / AS&IR (meaning) — *coordinate* on the REST layer (§1), *co-build* the ontology/domain layer.

- **Shared backlog** tied to Appendix B (Ontology Investigation, Studio Foundation, Agent Workflow POC, facilities/work-order correlation, lake/semantic feeds).
- **BU contributes** domain knowledge + stewardship authority (Documents 17, 18); **CogniBase contributes** the engine + OnBase depth.

Why co-development de-risks a lean vendor. A fair sponsor concern is key-person risk. Co-development is the answer: it **transfers operating knowledge to BU as the work proceeds**, so BU is never dependent on one person — while the core engine remains licensed, supported, and portable (open exit, plain-file artefacts). The model turns the small-vendor concern from a risk into a structured strength.

> *The engagement is designed to prove value in BU's own terms, on BU's own fabric, with a clean exit — the most credible posture for a partner to a "we build in-house" institution.*

CogniBase — Confidence, Human-in-the-Loop Review & Continuous Learning

Document 17 of 18 · BU-Aligned Library v2 · Renne Santiago

The adaptive trust loop: confidence thresholds, governed correction, and domain-by-domain maturation.

1. Why this document

Document 2 establishes a **static** discipline: a correlation is trustworthy only if a governed ontology sanctions it; everything else is *inferred and unverified.* This document makes that discipline **adaptive** — it adds a confidence-graded review queue and a governed learning loop so the system improves with use and **earns more autonomy in an area only after it has demonstrably learned that area's business.** The design borrows two proven precedents and corrects for where AI differs from them.

2. The precedents — and their limits

Precedent	Mechanism we borrow	Where AI differs (the correction)
OCR / ICR capture	A confidence score routes low-confidence items to a human verification queue; corrections feed back	OCR confidence is calibrated against pixels; **LLM self-confidence is not calibrated** — a model can be fluently confident and wrong
Translation memory / adaptive MT	Senior-translator post-edits become reusable "learned" assets	A bad learned phrase is cheap; a wrongly **learned relationship propagates fabrication at scale**

Conclusion: adopt the confidence-gated review + learning loop, but (a) gate on **evidence-based** confidence, not the model's self-assessment, and (b) make every learned item a **governed, steward-ratified proposal**, not an auto-absorbed fact.

3. The confidence model (evidence-based, not self-reported)

Confidence is a **composite of structural signals**, never the LLM's own "I'm sure":

- **Key strength** — was the entity resolved on a vetted canonical key, and how complete/unique is it?
- **Sanctioned-edge presence** — does a CrossLink/Normalizer authorize this join at all? (binary gate)
- **Source corroboration** — how many independent sources agree?
- **Grounding density** — citation count and retrieval score behind the claim.

These roll up into **bands**, each with a configurable threshold **per data area**:

Band	Meaning	Default disposition
Verified	Sanctioned edge + strong key + corroboration	Auto-accept, logged
Probable	Sanctioned edge, weaker evidence	Auto-accept above area threshold;

		else queue
Inferred	No sanctioned edge; pattern only	Never authoritative; labeled; queue for steward
Unsupported	Fails grounding	Withheld; refusal or "insufficient evidence"

4. The review queue & reviewers (tiered)

Items below an area's threshold enter a **review queue**. Reviewers are tiered so AI handles scale and humans hold ground truth:

Reviewer	Role	Authority
Inspector agent	Consistency & policy checks — does this contradict an existing sanctioned edge? violate a Gate?	Flags / blocks; cannot ratify new truth
Librarian agent	Provenance, deduplication, and placement — where does this belong, what does it cite, is it redundant?	Annotates / routes; cannot ratify
Domain steward (human)	Ratifies, rejects, or edits the proposed item	**Sole source of ground truth** until calibration is proven

AI agents **augment, never replace** the steward — avoiding an AI-certifies-AI loop where both are confidently wrong.

5. The learning loop (governed)

A correction is a **proposal**, not an immediate truth:

```
Flag/correction  →  Inspector+Librarian triage  →  Steward ratifies
   →  becomes a sanctioned item (Normalizer / CrossLink / annotation)
   →  carries provenance: who proposed, who ratified, when, on what evidence
   →  re-enters via the Hygiene Pipeline (Document 9), versioned
```

Deny-by-default is preserved: an un-ratified proposal changes nothing. This is the translation-memory idea **with governance bolted to its front** — it captures expert knowledge without letting a single mistaken or malicious correction teach the system a falsehood (**feedback-poisoning** defense).

6. Domain-by-domain maturation (changing the levels — safely)

Confidence thresholds are **per data area**, and they move with evidence:

1. **Cold start:** high review rate; most non-Verified items queued.

2. **Calibrate:** measure, per area, **human-agreement rate** and **audit pass rate** on reviewed items.

3. **Mature:** once an area's calibration clears a target over a sustained sample, **raise the threshold** so fewer items need review — the system has demonstrably *learned that business*.

4. **Monitor drift:** keep sampling; if agreement later drops, **auto-tighten** the threshold again. Maturation is reversible, never permanent.

This is the disciplined version of your "once the business is well-learned, change the confidence levels" — driven by measured calibration, not by feeling.

7. Per-area agents & scoped user feedback

- **Per-area agents** — a dedicated review agent per data domain (Financial Aid, Advancement, Facilities, HR…), each tuned to that area's entities, keys, and gates.
- **Scoped feedback** — users with **role/stewardship** in an area submit corrections that enter the governed loop. Authority is tiered, **not** equal to data access: *anyone with access may flag; a designated steward ratifies.* This mirrors the "senior translator" weighting — the people who own the business meaning are the ones whose corrections become truth.

8. Guardrails (the four rules)

1. **Evidence-based confidence only** — never gate on LLM self-confidence.

2. **Governed learning** — corrections are steward-ratified proposals; deny-by-default preserved.

3. **Authority by stewardship** — flag rights ≠ ratify rights.

4. **Data-driven, reversible thresholds** — raise on measured calibration; auto-tighten on drift.

9. Metrics (extends Document 2 §7 and Document 10)

- **Calibration error** per area — does "Probable" actually mean what its threshold claims?
- **Human-agreement rate** — % of agent/auto decisions a steward upholds.
- **Review-queue volume & SLA** — load and turnaround per area.
- **Learned items ratified vs. rejected** — and the rejection reasons (a healthy non-zero rejection rate proves the gate works).
- **Threshold-by-area + drift alerts** — current autonomy level per domain and any auto-tightening events.

10. Worked example (BU)

Financial-Aid document classification + enrollment correlation.

- *Cold start:* every non-Verified classification and every FA-doc⇄SIS correlation below threshold is queued. The Inspector agent checks each against existing sanctioned edges; the Librarian agent attaches provenance; the **FA office steward** ratifies.
- *Learning:* the steward corrects a recurring mis-classification; once ratified it becomes a sanctioned annotation with provenance and re-enters via hygiene.
- *Maturation:* after a sustained sample shows ≥ target human-agreement, CogniBase **raises the FA-area threshold** — routine items now auto-accept, freeing the steward for edge cases.
- *Drift:* a new FA form layout drops agreement; the monitor **auto-tightens** the threshold and re-queues until the steward teaches the new pattern.

11. Summary

Confidence-gated review turns CogniBase's trust layer from a fixed rule into a **system that learns a business and earns autonomy in it** — without ever surrendering the guarantee that a *believable* answer is also a *governed* one. It is your OCR/translation-memory instinct, corrected for the one way AI is different: **the machine's confidence is not evidence — the evidence is.**

> *Cross-references: Document 2 (ontology/trust), Document 3 (R15 confidence/feedback risk), Document 7 (feedback authority), Document 9 (learned items via hygiene), Document 10 (provenance & calibration in audit).*

CogniBase — BU Systems of Record: Affinity & Extension Map

Document 18 of 18 · BU-Aligned Library v2 · Renne Santiago

BU is not one database — it is a federation of domain systems of record. Where CogniBase can responsibly extend, and where it must not.

1. Why this document

The rest of this library focuses on OnBase, because that is CogniBase's home and BU's largest unstructured content store. But OnBase is one node in a wide federation: BU runs **dozens of authoritative systems of record**, each owning a domain. A sponsor evaluating CogniBase will rightly ask, *"This is about more than OnBase — how does it relate to our other systems, and which ones should it touch at all?"* This document answers that with three things: a **map** of BU's major systems of record, an **affinity model** of the entities they share, and an **inclusion test** that decides — under the same governed-correlation discipline as Document 2 — which extensions are valuable, which are policy-bounded, and which are forbidden.

The governing rule carries straight over: **shared subject matter is not permission to join.** High affinity between two systems is a reason to *consider* a sanctioned relationship, never to *assume* one.

2. BU's systems of record — two layers

BU's systems of record divide into two complementary layers: an **identity & persona backbone** that masters *who someone is*, and the **domain application systems** that master *what they do* in each area. CogniBase must resolve a person against the first before it dares correlate across the second.

2.1 Identity & persona backbone (the master layer)

System	Owner	Authoritative for
Person Identity	IS&T — Identity & Access Management (IAM)	The master **BU ID Number (BUID)** — the one canonical person key
Person Registry	IS&T — IAM	Person attributes synthesized across systems (including gender-affirming attributes)
Affiliates Database	IS&T — IAM	The **Affiliate persona** — uncompensated guests with validated access (vendors, volunteers, visiting researchers)
Analytical Services & Institutional Research (AS&IR)	AS&IR	The **source-of-record *rules*** — which system is authoritative for which demographic / bio-demographic attribute

The persona model. One human is addressed simultaneously through several **personas** — *Student* (MyBU / Registrar), *HR* (BUworks/SAP — active faculty/staff, emeriti, retirees), and *Affiliate* (Affiliates DB) — each mastered by a different system but unified by one **BUID**. This is the real-world reason CogniBase resolves people on the BUID and never on a name: *a vendor who is also an* *employee is one BUID wearing an Affiliate persona and an HR persona* — provable by identity, not inferable by string match (cf. Document 2, Example 1).

AS&IR is BU's existing source-of-record authority. It already defines which system is authoritative for which attribute. CogniBase's Normalizers and CrossLinks should **inherit and defer to AS&IR's source-of-record rules**, and AS&IR — with domain stewards — is the natural ratifier of CogniBase's sanctioned relationships (Document 17). BU already has the governance body the trust layer needs.

2.2 Domain application systems of record

Domain	System(s) of record	Role
Student records / enrollment	**MyBU Student** — Oracle PeopleSoft *Campus Solutions* (replaced the 40-year mainframe SIS / Student Link; ~11M records migrated)	Admissions, financial aid, records, curriculum, financials, advising
HR / Finance / Procurement	**BUworks** — *SAP* (Finance, HCM/HR, Procurement, Reporting); MyBUworks UI	Employee, payroll, vendor, purchasing, GL
Admissions	**Slate** (+ Salesforce applicant, Liaison)	Inquiry → matriculation
Housing / residence	**StarRez**	Housing applications, assignments
Advancement / alumni / donors	**Blackbaud CRM** (system of record) + iModules Encompass + EverTrue	360° constituent: biographical, gifts, education, scholarships, events
Research administration	**Kuali Research** (proposals/awards; Negotiations for industry contracts); **Huron** Grants & Agreements + COI; **INSPIR II** (IRB human-subjects)	Grants, agreements, compliance — *Kuali interfaces with SAP*
Facilities / campus operations	**CAMMS** (maintenance & work-order central DB); **PMWeb** (capital projects); **25Live Pro** (room/event scheduling); **FMS: Workplace** (space)	Buildings, assets, work orders, space, projects
Library / archives	**Ex Libris Alma** (library management); **ArchivesSpace** (finding aids); **BU Digital Library**; Gotlieb Archival Research Center (University Archives)	Holdings, archival description, digitized heritage
IT service management	**ServiceNow** (primary ITSM; the TechWeb portal)	Tickets, knowledge, service catalog
Document / content management (ECM)	**Hyland OnBase** — *integrates across student, advancement, HR/finance, and research-admin systems*	The cross-cutting document layer — **CogniBase's home**
Learning	**Blackboard** (LMS); Course Catalog / Bulletin	Courses, sections, content
Other domain systems	WordPress (web/CMS), CS Gold (campus card/access), Parchment (transcripts), TerraDotta (study abroad), Handshake (careers), Accommodate (disability)	Domain-specific
Research computing	**Research Computing Infrastructure / Management Systems** (IS&T Research Computing)	HPC access, account provisioning, research-environment authorizations
Professional / non-degree	**Destiny One** (Center for Professional Education)	Independent SoR for non-degree professional enrollments & credentials
Authentication	**Entra ID** (Azure AD) / Kerberos	Sign-on — the *identity master* is

		Person Identity / BUID (see §2.1)
Legal / General Counsel	*No single contract-management system identified publicly.* Contracts appear distributed across **Kuali Negotiations** (research), **SAP** (procurement), and **OnBase** (document storage) — **to be confirmed with BU.**	Contracts, legal records (privileged)

Note on accuracy: this map is compiled from BU's public IS&T/TechWeb materials and the AIDA dossier; it should be ratified against BU's own application portfolio before any pilot scopes a system. The legal/contract layer in particular is an open question to confirm with the Office of the General Counsel.

3. OnBase is already the cross-cutting seam

The single most important fact for CogniBase's extension story: **OnBase is not a domain silo — BU already runs it across student, advancement, HR/finance, and research-administration systems** as the shared document layer. CogniBase, which masters OnBase, therefore *already sits at the seam* between these systems of record. Extending CogniBase to correlate across them is not a leap into unrelated territory — it is following document relationships OnBase already maintains, and making them governed, semantic, and auditable.

4. The affinity model — the entities these systems share

Systems of record connect through a small set of **shared entities**. These are the *only* legitimate join surfaces; everything else is coincidence.

Shared entity	Appears in	Sanctioned join key
Person (universal)	MyBU (student), SAP (employee), Slate (applicant), Blackbaud (donor/alum), Kuali/Huron (PI), StarRez (resident), **Affiliates DB** (vendor/guest), CS Gold (cardholder)	**BUID** — mastered by **Person Identity** (IAM), distributed with 400+ attributes via the Person Broker/CDM — *never name or email*
Space / Building / Asset	CAMMS, PMWeb, 25Live, FMS: Workplace, OnBase facilities docs	Facilities asset / building / space ID
Course / Curriculum / Section	MyBU, Blackboard, Course Catalog	Course/section ID
Award / Grant / Protocol	Kuali, Huron, INSPIR, SAP	Award / protocol number
Constituent / Gift / Fund	Blackbaud, SAP financials, scholarships (→ student)	Constituent / fund ID
Document (overlay)	OnBase across all of the above	DocType + keyword → canonical entity

Each system maintains its **own ID space.** "Person" in MyBU and "Person" in Blackbaud are the *same human* only when a validated identifier says so — the exact entity-resolution discipline of Document 2. This is why affinity is an invitation to model a relationship, not a license to assume one.

5. The inclusion test — which systems CogniBase should touch

Affinity is graded into three tiers, applied through the access model (Document 7) and the trust layer (Document 2):

Tier A — Natural synergy (governed correlation is valuable)

OnBase documents ⇄ **MyBU** (student records), ⇄ **SAP** (AP/procurement), ⇄ **Kuali/Huron** (grant files); the **facilities cluster** (CAMMS ⇄ PMWeb ⇄ 25Live ⇄ FMS) ⇄ OnBase facilities docs. These share clean entities and clear business questions ("show invoices over $10k stuck in approval," "which facilities work orders have signed-off completion docs"). **Start here.**

Tier B — Policy- and ethics-bounded (possible, but gated)

Advancement (Blackbaud) ⇄ student records, and **admissions ⇄ enrolled records.** The entities relate, but FERPA and ethical-fundraising boundaries apply — a donor's gift history must not leak into a student-services answer, and vice versa. Permitted *only* through an explicit Gate with a recorded agreement and a domain steward (Documents 7, 17).

Tier C — Excluded / out of scope

INSPIR / IRB human-subjects, **Student Health** (HIPAA — CogniBase does not ingest HIPAA data), **Accommodate** (disability), and **privileged legal records** (attorney-client privilege). These are deliberately outside CogniBase's reach; high affinity does not override regulatory or privilege boundaries.

> *The inclusion test is the federation-scale version of the trust thesis: the more systems are in play, the more tempting a fluent-but-false cross-system analysis becomes, and the more important it is that **every cross-system join is sanctioned, keyed, gated, and audited.***

6. Phased extension (sponsor-friendly)

1. **OnBase-centric (pilot):** prove governed correlation *within* OnBase and to one adjacent Tier-A system (e.g., MyBU).

2. **Tier-A federation:** add SAP and the facilities cluster as sanctioned, sponsored use cases arise.

3. **Semantic feed:** contribute resolved entities and sanctioned relationships into BU's Lexicon / enterprise knowledge graph (Document 15).

4. **Tier-B only on demand:** never speculatively — only with a sponsor, a steward, a Gate, and a recorded agreement.

Each phase is gated on a *named sponsor and a real question*, never on technical possibility alone.

7. Summary

BU is a federation of domain systems of record — MyBU/Campus Solutions for students, SAP for finance and HR, CAMMS and its siblings for facilities, Alma and ArchivesSpace for the library and archives, Kuali and Huron for research, Blackbaud for advancement, ServiceNow for IT — with **OnBase already threaded across many of them** as the document layer. CogniBase's opportunity is to turn those existing document relationships into **governed, semantic, auditable correlations** — and its discipline is knowing that **the entities these systems share are an invitation to model a relationship, never a permission to assume one.** That is what lets CogniBase scale across BU's systems without scaling the risk of fabricated insight.

> *Cross-references: Document 2 (governed correlation), Document 7 (Gates/access), Document 15 (lake & knowledge graph), Document 17 (confidence & stewardship).*

CogniBase — Appendix A: MapSnap × CogniBase Tandem & Pre-Ingestion Augmentation

Appendix A · BU-Aligned Library v2 · Renne Santiago

How the discovery family (MapSnap, AutoPDF) can feed and strengthen CogniBase — including a forward-looking option to pre-align OnBase content at ingestion. Marked optional / exploratory; not required for the core pilot.

A.1 Why this appendix

CogniBase does not stand alone. It is the OnBase apex of a small **discovery family** that shares one engine: **MapSnap** (the schema-intelligence ancestor that understands *any* enterprise database) and **AutoPDF** (a local-first document-capture toolkit that already emits an OnBase XML Index DIP). This appendix sketches how the three evolve in tandem, and explores a deliberately speculative idea: using AutoPDF + MapSnap to **normalize and augment documents with metadata *before* they enter OnBase**, so OnBase content is born pre-aligned with the rest of BU's data.

A.2 MapSnap × CogniBase — a division of labor

	MapSnap	CogniBase
Domain	*Structured* systems of record (SAP/BUworks, MyBU/Campus Solutions, CAMMS, StarRez…)	OnBase (unstructured/document) + cross-source correlation
Core skill	Schema mapping, soft-FK inference, NL→SQL, value catalogs	Governed correlation, RAG, Normalizers/Gates/Federator
Shared engine	The same discovery + enrichment pipeline runs in both	

The tandem. MapSnap is the natural **onboarding tool for the federation in Document 18.** Pointed at each structured system of record, it reads the schema and **proposes canonical-key mappings** — e.g., "SAP vendor key ≡ canonical `PersonTaxID`," "CAMMS asset key ≡ canonical `AssetID`." Those proposals become **candidate Normalizers** that CogniBase consumes and a steward ratifies (Document 17). MapSnap accelerates building the ontology that CogniBase governs; CogniBase brings OnBase and the correlation discipline MapSnap doesn't have. They are two ends of one pipeline: *understand the structured sources → correlate them with the documents.*

A.3 The anterior phase — AutoPDF + MapSnap pre-ingestion normalization

Today, documents enter OnBase and CogniBase reconstructs their canonical keys *after the fact*. The proposal is to move some of that work **upstream, to capture time, where context is richest:**

```
Incoming documents
   → AutoPDF        : split · OCR/extract fields · anchor-relative mapping → metadata index
   → MapSnap map    : resolve extracted fields to canonical entities/keys using the
                      value catalogs MapSnap built from the structured systems of record
   → Normalized, entity-tagged metadata (canonical keys + confidence + provenance)
   → OnBase ingestion (DIP)
```

Net effect: a document enters OnBase already carrying, where evidence supports it, a resolved `BUID`, a canonical entity tag, a confidence score, and a provenance stamp — so downstream CogniBase correlation is *pre-aligned* rather than rebuilt. AutoPDF already produces OnBase-ready DIP output; MapSnap supplies the canonical vocabulary; CogniBase later consumes the result under governance.

A.4 The augmentation-keyword idea (the long shot) — two variants

Goal: attach extra metadata to OnBase documents that is *invisible to standard OnBase retrieval* but *accessible to CogniBase*, pre-aligning OnBase with other sources. Two ways to do it:

Variant 1 — CogniBase overlay (RECOMMENDED). Store the augmentation (canonical keys, confidence, provenance) in a **CogniBase-side store keyed by the OnBase Document Handle.** OnBase stays pristine — no config change, read-mostly preserved, fully reversible — and CogniBase joins augmentation to documents on the handle. Cleanest, lowest-risk, no OnBase-governance friction.

Variant 2 — dedicated "augmentation" KeywordTypes (OPTIONAL). Add purpose-built KeywordTypes (e.g., `CB_CanonicalEntity`, `CB_BUID`, `CB_Confidence`, `CB_Provenance`) to DocTypes, assigned but kept **out of the standard retrieval / Custom Query surfaces** via display and security configuration. The keys then travel *inside* OnBase; CogniBase reads them via the API. Requires OnBase configuration **and write access** — a deliberate expansion of CogniBase's normally read-mostly posture.

Plausibility (honest read). Variant 1 is straightforwardly plausible and low-risk — do this first. Variant 2 is technically feasible (OnBase lets you add KeywordTypes and limit their exposure), but **"invisible" is a configuration/security posture, not a truly hidden field**: OnBase admins and auditors can still see these keywords — which is *correct* (nothing should be genuinely secret), but it means the value is "kept out of the everyday retrieval UI," not "undetectable." It also touches OnBase indexing and config and needs write access. Use Variant 2 only if BU specifically wants the metadata embedded in OnBase and accepts that cost.

A.5 Guardrails (non-negotiable)

1. **Augmentation is a *claim*, not a fact.** A resolved `BUID` on an invoice is an entity-resolution assertion — it must carry **confidence + provenance**, and a low-confidence resolution stays *inferred* and queued for a steward (Documents 2, 17). **Never pre-assert a join you cannot evidence** — baking a false correlation into the system of record at ingestion is worse than producing one at query time.

2. **No ACL / privacy bypass.** "Invisible to OnBase retrieval but visible to CogniBase" must **not** become a channel that smuggles data past OnBase access controls. Augmentation keywords carry only **non-sensitive canonical tags/keys/confidence — never new PII** — and CogniBase access to them still honors identity and Gates. *(A security reviewer will probe exactly this; the honest answer is the overlay variant and a strict no-new-PII rule.)*

3. **Reversibility.** The overlay is trivially reversible; any in-OnBase keyword must be removable.

4. **Read-mostly stays the default.** Writing to OnBase is an explicit, governed exception — never the standing posture.

A.6 Phasing & value

This is **Phase 0 / optional** — it accelerates the later federation (Document 18) but is **not** required for the OnBase pilot. Start with the overlay (Variant 1); consider in-OnBase augmentation keywords only on explicit BU request. The value is twofold: it turns the discovery family into one coherent pipeline — **capture (AutoPDF) → understand (MapSnap) → align (augmentation) → correlate (CogniBase)** — and it pre-pays the entity-resolution cost at ingestion, where the document's own context makes resolution most reliable.

A.7 Summary

MapSnap and AutoPDF are not separate products bolted onto CogniBase — they are the **front half of the same governed pipeline.** MapSnap onboards the structured systems of record and proposes the canonical keys CogniBase governs; AutoPDF can pre-normalize documents at capture; and, optionally, a thin augmentation layer can let OnBase content arrive pre-aligned with the rest of BU's data — provided every augmentation is evidenced, privacy-safe, and reversible. It is the discovery family doing at ingestion, under governance, what CogniBase otherwise does at query time.

> *Cross-references: Document 2 (governed correlation), Document 17 (confidence & stewardship), Document 18 (the systems-of-record federation).*

Appendix A · Frame A · CogniBase BU-Aligned Library v2 · generated 2026-06-28

CogniBase — Appendix B: Alignment to BU's 2026 Roadmap (Four Lanes)

Appendix B · BU-Aligned Library v2 · Renne Santiago

CogniBase is not a net-new initiative competing for budget — it advances items already on BU's funded 2026 roadmap. This appendix crosswalks CogniBase to each lane.

B.1 Why this appendix

The strongest argument to a sponsor is not "fund something new" — it is **"fund something that accelerates work you have already prioritized."** BU's 2026 roadmap is published in four lanes: **AI Engineering, Data Engineering, Enterprise Architecture, and Essential Services.** This appendix maps CogniBase onto specific line items in each, with an honest read of where the fit is strong and where it is only supporting.

> *Source note: roadmap items are taken from BU's published AI & Data Engineering roadmap (four-lane crawl, 2026-06-27). The live roadmap is a tabbed, evolving page — these mappings should be re-confirmed against it before any proposal.*

B.2 Fit at a glance

Lane	CogniBase fit	Why
AI Engineering	★★★ Strongest	"Ontology Investigation" is practically a CogniBase brief; MCP-native; document AI
Data Engineering	★★★ Strong	Lake ingestion + semantic layer + catalog + profiling all have CogniBase/MapSnap roles
Enterprise Architecture	★★ Strong & concrete	Work-order/asset, SIS/Jenzabar/mainframe migration, Huron, OpenClaw, IAM
Essential Services	★ Supporting only	Legacy ADW/BI/DBA ops — CogniBase assists migrations, isn't central (stated honestly)

B.3 Lane 1 — AI Engineering (strongest fit)

Roadmap item	How CogniBase advances it	Library ref
Ontology Investigation	CogniBase's governed ontology — Normalizers/CrossLinks, sanctioned correlation, knowledge-graph feed — *is* an ontology investigation, grounded in OnBase + the systems federation	Docs 2, 15, 18
TerrierAI Assistant Pilot (OpenClaw)	CogniBase registers as an MCP tool the OpenClaw-based Assistant calls for OnBase questions	Doc 11
Migrate to MCP Gateway / MCP Gateway (infra)	CogniBase publishes to the `mcp-gateway-registry`; native, governed tool surface	Docs 11, 14

TerrierAI Studio (document/data processing)	Document intelligence + RAG over OnBase with citations and audit	Docs 5, 10
AI Code Generation – Data Engineering / Claude Code	MapSnap + CogniBase schema discovery accelerates pipeline/code generation	Appendix A
Consulting Partner AI Landscape/Roadmap Review	The named, RFP-shaped entry point for this engagement	Doc 16

B.4 Lane 2 — Data Engineering (strong fit)

Roadmap item	How CogniBase advances it	Library ref
Lake ingestion targets (Campus Solutions/SIS, SAP, Kuali, Huron, Blackbaud, StarRez, 25Live, Slate, Destiny One…)	CogniBase emits curated OnBase entities to Cortex; **MapSnap onboards the structured sources and proposes canonical keys**	Docs 15, 18; Appendix A
Automated xForm Processing · Data Profiling · Data Validation	AutoPDF capture/extract + CogniBase hygiene pipeline (profiling, PII detection, validation)	Doc 9; Appendix A
Insights: Cube Semantic Layer POC · Strategic KPIs Layer	CogniBase's sanctioned relationships & canonical terms feed Lexicon/Cube	Doc 15
Infrastructure: OpenMetadata Catalog	CogniBase provenance/lineage aligns with the catalog; learned items carry source + steward	Doc 10
TerrierData Connect (move off SnapLogic) · BU Message Bus	Not CogniBase's job — but its open-standards, event-friendly posture complements rather than competes	Docs 4, 13

B.5 Lane 3 — Enterprise Architecture (strong, concrete fit)

Roadmap item	How CogniBase advances it	Library ref
Work Order & Asset Management · FacilitiesOS	Governed correlation across the facilities cluster (CAMMS/PMWeb/25Live/FMS) + OnBase facilities docs	Doc 18
SIS Stabilization · Jenzabar Retirement · Mainframe Archive	MapSnap schema archaeology turns undocumented legacy schemas into plain-English, join-aware maps for migration	Appendix A
Huron IRB · Huron Grants	Research-admin document correlation — Tier-B/C bounded (gated; IRB excluded)	Doc 18
OpenClaw AI Assistant	CogniBase as the governed OnBase tool behind the Assistant	Doc 11
RightCrowd PIAM · SIS IAM	Identity alignment on **BUID / Entra ID**; CogniBase honors OnBase ACLs and the persona model	Docs 7, 18
(OnBase — the ECM beneath many of these)	CogniBase's home; the layer that makes OnBase content institutional knowledge	All

B.6 Lane 4 — Essential Services (supporting only — stated honestly)

This lane is traditional data-warehousing, BI-platform, and database-administration operations (ADW enhancements, MicroStrategy migration/re-licensing, *Evaluate CoPilot for Power BI*, mainframe→SQL/Coldstore archive, SQL Server upgrades). **CogniBase is not central here**, and the document should not pretend otherwise. Its only honest roles are:

- **Migration assistance** — MapSnap mapping legacy ADW / MicroStrategy / mainframe schemas during rehost and archive work.
- **A forward direction** — CogniBase's governed semantic layer is the modern, AI-native counterpart to the legacy BI stack BU is migrating — a *destination*, not a like-for-like replacement.

Claiming more than this in Lane 4 would be exactly the kind of overstated affinity the rest of this library argues against.

B.7 Summary

CogniBase maps **most strongly to AI Engineering** (where *Ontology Investigation* is practically its charter), **concretely to Data Engineering and Enterprise Architecture** (lake ingestion, semantic layer, facilities/asset, legacy-migration archaeology, OpenClaw, IAM), and is **supporting-only in Essential Services.** The sponsor's takeaway: funding CogniBase advances roughly a dozen items *already on BU's prioritized roadmap* — it is an accelerator of the plan, not an addition to it.

B.8 A tandem-development reading

Because these are *shared* objectives, the natural way to pursue them is **together.** The lane items above double as a **co-development backlog** — Ontology Investigation, Studio Foundation, Agent Workflow POC, the facilities/work-order correlation, and the lake/semantic feeds are all candidates for a governed tandem build in which **BU owns the institution-specific ontology and CogniBase contributes the licensed engine.** See **Document 16 §7 (Co-Development Model)** for the ownership split and how co-development de-risks a lean vendor through skills transfer.

> *Cross-references: Document 1 (vision alignment), Document 16 (pilot, engagement & co-development), Document 18 (systems federation), Appendix A (MapSnap/AutoPDF tandem).*